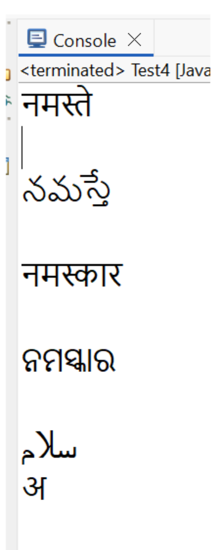
Character Literal Contd..

```java
package com.ravi.character_literal;

//Java supports UNICODE
public class Test4
{
    void main()
    {
        String hindi = "\u0928\u092E\u0938\u094D\u0924\u0947";
        IO.println(hindi);


        String telugu = "\u0C28\u0C2E\u0C38\u0C4D\u0C24\u0C47";
        IO.println("\n"+telugu);

        String marathi = "\u0928\u092E\u0938\u094D\u0915\u093E\u0930";
        IO.println("\n"+marathi);

        String odia = "\u0B28\u0B2E\u0B38\u0B4D\u0B15\u0B3E\u0B30";
        IO.println("\n"+odia);

        String urdu = "\u0633\u0644\u0627\u0645";
        IO.println("\n"+urdu);


        IO.println("\u0905");


    }
}
```




```java
Test5.java
-------------
void main()
{
    int x = 'A';
  int y = 'B';

    IO.println(x+y); //131

    IO.println('A' + 'A'); // 130 [Numeric Promotion]

    IO.println("A"+"A"); //AA [String concatenation]
}
```

```java
 Test6.java
 ------------
 //Range of UNICODE Value (0 - 65535) OR '\u0000' to  '\uffff'
 void main()
 {
          char ch1 =  65535;
          IO.println("ch value is :"+ch1);

          char ch2 = 65536;
          IO.println("ch value is :"+ch2);
 }
```

```java
 Test7.java
 --------------
 //WAP in java to describe unicode representation of char in hexadecimal format

  void main()
      {
           int ch1 = '\u0000';
           IO.println(ch1);

           int ch2 = '\uffff';
           IO.println(ch2);

           char ch3 = '\u0041';
       IO.println(ch3);  //A

           char ch4 = '\u0061';
           IO.println(ch4); //a
      }
```

```java
 Test8.java
 ------------
 void main()
 {
     char c1 = 'A';
     char c2 = 65;
     char c3 = '\u0041';
     IO.println("c1 = "+c1+", c2 ="+c2+", c3 ="+c3);
 }
```

```java
 Test9.java
 ------------
 void main()
 {
     int x = 'A';
     int y = '\u0041';
     IO.println("x = "+x+" y ="+y);
 }
```

```java
Test10.java
--------------
//Every escape sequence is char literal

void main()
{
          char ch ='\n';
          IO.print("Hello");
          IO.println(ch);

          IO.println("'Hello Java'");  //'Hello Java'
          IO.println("\"Hello learner!!!!\"");

}
```

```java
Test11.java
--------------
 void main()
 {
  IO.println("Character size is :"+Character.SIZE);
  IO.println("Minimum value of Character :"+(int)Character.MIN_VALUE);
  IO.println("Maximum value of Character :"+(int)Character.MAX_VALUE);

     int c1 = '\u0000';
     int c2 = '\uffff';
     IO.println(c1+" : "+c2);
 }
```

Boolean Literal :
----

* It is used to represent two states i.e true OR false.

* We have only one data type i.e boolean data type which can store **1 bit of information and the size of boolean data type depends upon JVM implementation.**
  Example :
```java
              boolean isEmpty = true;
              boolean isValid = false;
```

* Unlike C & C++, we **cannot assign 0 and 1 to boolean data type becaue in java 0 & 1 represents integral literals.**
  Example :
```java
              boolean isValid = 0;  //Error [Valid in C & C++ but invalid in Java]
```

* String value we cannot assign to boolean type.
  Example :
```java
              boolean b = "true"; //error
```

```java
  Test1.java
  ------------
  void main()
  {
     boolean isValid = true;
     boolean isEmpty = false;

     IO.println(isValid);
     IO.println(isEmpty);
  }
```

```java
  Test2.java
  ------------
  void main()
  {
     boolean c = 0;
     boolean d = 1;
     IO.println(c);
     IO.println(d);
   }
```

```java
  Test3.java
  ------------
  void main()
  {
       boolean isValid = "true";
       IO.println(isValid);
  }
```

String Literal :

String is a predefined class available in java.lang (Library) Package.

String is a collection of alpha-numeric character which is enclosed by double quotes. These characters can be alphabets, numbers, symbol or any special character.
Example:
"India"
"Hyderabad"
"B-61, S R Nagar"

In java we can create String object by using following 3 ways:

1) By using String Literal
   String str1 = "india";

2) By using new keyword
   String str2 = new String("Hyderabad");

3) By using Character array [Old Technique]
   char ch[] = {'R', 'A', 'J'};
   String s3 = new String(ch);

//Programs :

```java
 Test1.java
 ------------
 //String is collection of alpha-numeric character

  void main()
      {
           String x = "B-61 Hyderabad";
           IO.println(x);

           String y = "123";
           IO.println(y);

           String z = "67.90";
           IO.println(z);

           String p = "A";
           IO.println(p);
      }
```

```java
Test2.java
------------
//IQ
void main()
{
     String s = 15 + 29 +"Ravi"+40 + 40;
     IO.println(s);
}
```

```java
 Test3.java
 -------------
 //Three Ways to create the String Object

 void main()
 {
      //By using Literal
      String s1 = "Java";
      IO.println(s1);

    //By using new keyword
      String s2 = new String("Hyderabad");
      IO.println(s2);

    //Character Array
    char ch[] = {'R', 'A', 'J'};
      String s3 = new String(ch);
      IO.println(s3);
 }
```

---

4) Punctuators:

It is also called separators.

It is used to inform the compiler how things are grouped in the code.

() {} [] ; , . @